2.1 In each of the following find the pdf of Y.

a) $Y = X^3$ and $f_X(x) = 42x^5(1-x), 0 < x < 1$

Solution: $f_X(x) = 42x^5(1-x), 0 < x < 1; y = x^3 = g(x)$, monotone and $y \in (0,1)$. Use Theorem 2.1.5

$$\begin{aligned} f_Y(y) &= f_x(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| \\ &= f_x(y^{1/3})\frac{d}{dy}(y^{1/3}) \\ &= 42y^{5/3}(1-y^{1/3})(\frac{1}{3}y^{-2/3}) \\ &= 14y(1-y^{1/3}) \\ &= 14y - 14y^{4/3}, 0 < y < 1. \end{aligned}$$

b) $Y = 4X + 3$ and $f_X(x) = 7e^{-7x}, 0 < x < \infty$

Solution: $f_X(x) = 7e^{-7x}, 0 < x < \infty; y = 4x + 3$, monotone and $y \in (3, \infty)$. Use Theorem 2.1.5.

$$\begin{aligned} f_Y(y) &= f_X(\frac{y-3}{4})\left|\frac{d}{dy}(\frac{y-3}{4})\right| \\ &= 7e^{-(7/4)(y-3)}\left|\frac{1}{4}\right| \\ &= \frac{7}{4}e^{-(7/4)(y-3)}, 3 < y < \infty. \end{aligned}$$

c) $Y = X^2$ and $f_X(x) = 30x^2(1-x)^2, 0 < x < 1$

Solution: $F_Y(y) = P(0 \le X \le \sqrt{y}) = F_X(\sqrt{y})$. Then $f_Y(y) = \frac{1}{2\sqrt{y}}f_X(\sqrt{y})$. Therefore

$$\begin{aligned} f_Y(y) &= \frac{1}{2\sqrt{y}}30(\sqrt{y})^2(1-\sqrt{y})^2 \\ &= 15y^{\frac{1}{2}}(1-\sqrt{y})^2, 0 < y < 1. \end{aligned}$$

2.2 In each of the following find the pdf of Y. Use Theorem 2.1.5 to find solutions.

a) $Y = X^2$ and $f_X(x) = 1, 0 < x < 1$

Solution: Let $g^{-1}(y) = \sqrt{y}$ and $\frac{d}{dy}g^{-1}(y) = \frac{1}{2}y^{-\frac{1}{2}}$. Then,

$$\begin{aligned} f_Y(y) &= f_x(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| \\ &= f_x(y^{\frac{1}{2}})\frac{d}{dy}y^{1/2} \\ &= \frac{1}{2}y^{-\frac{1}{2}}, 0 < y < 1. \end{aligned}$$

b) $Y = -\log X$ and X has pdf $f_X(x) = \frac{(n+m+1)!}{n!m!}x^n(1-x)^m, 0 < x < 1, m, n,$ positive integers.

Solution: Let $g^{-1}(y) = e^{-y}$ and $\frac{d}{dy}g^{-1}(y) = -e^{-y}$. Then,

$$\begin{aligned} f_Y(y) &= f_x(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| \\ &= f_x(e^{-y})\frac{d}{dy}e^{-y} \\ &= \frac{(n+m+1)!}{n!m!}e^{-y(n+1)}(1-e^{-y})^m e^{-y}, 0 < y < \infty. \end{aligned}$$

c) $Y = e^X$ and X has pdf $f_X(x) = \frac{1}{\sigma^2}xe^{-(x/\sigma)^2/2}/, 0 < x < \infty, \sigma^2$ a positive constant.

Solution: Let $g^{-1}(y) = logy$ and $\frac{d}{dy}g^{-1}(y) = \frac{1}{y}$. Then,

$$\begin{aligned} f_Y(y) &= f_x(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| \\ &= f_x(\log y)\frac{d}{dy}\log y \\ &= \frac{1}{\sigma^2}\log y e^{-((\log y)/\sigma)^2/2}\frac{1}{y} \\ &= \frac{1}{\sigma^2}\frac{\log y}{y}e^{-((\log y)/\sigma)^2/2}, 1 < y < \infty. \end{aligned}$$

2.3 Suppose X has the geometric pmf $f_X(x) = \frac{1}{3}\left(\frac{2}{3}\right)^x, x = 0, 1, 2, \ldots$ Determine the probability distribution of $Y = \frac{X}{X+1}$. Note that here both X and Y are discrete random variables. To specify the probability distribution of Y, specify its pmf.

Solution:

$$\begin{aligned}
P(Y=y) &= P(\frac{X}{X+1}=y) \\
&= P(X=y(X+1)) \\
&= P(X-Xy=y) \\
&= P(X(1-y)=y) \\
&= P(X=\frac{y}{1-y}) \\
&= \frac{1}{3}(\frac{2}{3})^{y/(1-y)}
\end{aligned}$$

where $y=0,\frac{1}{2},\frac{2}{3},\frac{3}{4},\ldots,\frac{x}{x+1},\cdots$

2.6 In each of the following find the pdf of Y.

a) $f_X(x)=\frac{1}{2}e^{-|x|}, -\infty<x<\infty; Y=|X|^3$

Solution: Use Theorem 2.1.8 to solve this problem. First, let $A_0=\{0\}, A_1=(-\infty,0)$ and $A_2=(0,\infty)$.
CASE A_1:

$$\begin{aligned}
y &= |x|^3 = -x^3 \\
-y &= x^3 \\
(-y)^{1/3} &= x \\
\left|\frac{dx}{dy}\right| &= \left|\frac{1}{3}(-y)^{-2/3}(-1)\right| = \frac{1}{3}y^{-2/3}
\end{aligned}$$

CASE A_2:

$$\begin{aligned}
y &= |x|^3 = x^3 \\
y &= x^3 \\
y^{1/3} &= x \\
\left|\frac{dx}{dy}\right| &= \frac{1}{3}(-y)^{-2/3}
\end{aligned}$$

Use Theorem 2.1.8 to obtain the pdf of Y:

$$\begin{aligned}
f_Y(y) &= \frac{1}{2}e^{-|y^{1/3}|}\frac{1}{3}y^{-2/3}+\frac{1}{2}e^{-|(-y)^{1/3}|}\frac{1}{3}y^{-2/3} \\
&= \frac{1}{6}e^{-|y^{1/3}|}y^{-2/3}+\frac{1}{6}e^{-|y^{1/3}|}y^{-2/3} \\
&= \frac{1}{3}e^{-y^{1/3}}y^{-2/3}, 0\le y<\infty
\end{aligned}$$

b) $f_X(x) = \frac{3}{8}(x+1)^2, -1 < x < 1; Y = 1 - X^2$

Solution: Let $A_0 = \{0\}, A_1 = (-1,0)$ and $A_2 = (0,1)$. Then $g_1 = 1 - x^2$ on A_1 and $g_2 = 1 - x^2$ on A_2. Now, $g_z{}^{-1}(y) = \sqrt{1-y}$ with derivative $-\frac{1}{2}(1-y)^{-1/2}$. Use Theorem 2.1.8 to obtain the pdf of Y for $y \in (0,1)$:

$$
\begin{aligned}
f_Y(y) &= \frac{3}{8}(-(1-y)^{1/2}+1)^2\frac{1}{2}(1-y)^{-1/2} + \frac{3}{8}((1-y)^{1/2}+1)^2\frac{1}{2}(1-y)^{-1/2} \\
&= \frac{1}{2}\frac{3}{8}(1-2(1-y)^{1/2}+(1-y))(1-y)^{-1/2} + \frac{1}{2}\frac{3}{8}((1-y)+2(1-y)^{1/2}+1)(1-y)^{-1/2} \\
&= \frac{1}{2}\frac{3}{8}((1-y)^{-1/2}-2+(1-y)^{1/2}) + \frac{1}{2}\frac{3}{8}((1-y)^{1/2}+2+(1-y)^{-1/2}) \\
&= \frac{3}{8}(1-y)^{-1/2} + \frac{3}{8}(1-y)^{1/2}
\end{aligned}
$$

c) $f_X(x) = \frac{3}{8}(x+1)^2, -1 < x < 1; Y = 1 - X^2$ if $x \leq 0$ and $Y = 1 - X$ if $X \geq 0$

Solution: Let $A_0 = \{0\}, A_1 = (-1,0)$ and $A_2 = (0,1)$. Then $g_1(x) = 1 - x^2$ on A_1 and $g_1(x) = 1 - x$ on A_2. Now $g_1(x) = 1 - x^2$ and $g_2(x) = 1 - x$ on A_2. Now, $g_1{}^{-1}(y) = -\sqrt{1-y}$ and $g_2{}^{-1}(y) = 1 - y$, with derivatives $-\frac{1}{2}(1-y)^{-1/2}$ and -1, respectively. Use Theorem 2.1.8 to obtain the pdf of Y:

$$
\begin{aligned}
f_y(y) = &\frac{3}{8}(-(1-y)^{1/2}+1)^2\frac{1}{2}(1-y)^{-1/2} + \frac{3}{8}((1-y)+1)^2 + \\
&\frac{3}{16}(1-(1-y)^{1/2})^2\frac{1}{(1-y)^{1/2}} + \frac{3}{8}(2-y)^2, y \in (0,1)
\end{aligned}
$$

2.7 (a.)

For $-1 < x < 1 \Rightarrow 0 < y < 1$ use thm 2.1.8

$f_X(x) = \frac{2}{9}(x+1) \quad Y = X^2$

$-1 < x < 0$

$x = -\sqrt{y}$

$\left|\frac{dx}{dy}\right| = \frac{1}{2\sqrt{y}}$

$0 < x < 1$

$x = \sqrt{y}$

$\left|\frac{dx}{dy}\right| = \frac{1}{2\sqrt{y}}$

$$f_Y(y) = \frac{2}{9}(-\sqrt{y}+1)\frac{1}{2\sqrt{y}} + \frac{2}{9}(\sqrt{y}+1)\frac{1}{2\sqrt{y}} = \frac{1}{9\sqrt{y}}(-\sqrt{y}+1+\sqrt{y}+1)$$
$$= \frac{2}{9\sqrt{y}}$$

For $1 < x < 2 \Rightarrow 1 < y < 4$ do monotone transformation

$$f_Y(y) = \frac{2}{9}(\sqrt{y}+1)\frac{1}{2\sqrt{y}} = \frac{\sqrt{y}+1}{9\sqrt{y}} = \frac{1}{9} + \frac{1}{9\sqrt{y}}$$

So

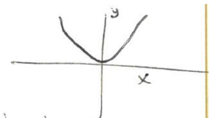
$$\boxed{f_Y(y) = \begin{cases} \frac{2}{9\sqrt{y}} & 0 < y < 1 \\ \frac{1}{9} + \frac{1}{9\sqrt{y}} & 1 < y < 4 \\ 0 & \text{otherwise} \end{cases}}$$

2.12 We have $\tan x = y/d$, therefore $\tan^{-1}(y/d) = x$ and $\frac{d}{dy}\tan^{-1}(y/d) = \frac{1}{1+(y/d)^2}\frac{1}{d}dy = dx$. Thus,

$$f_Y(y) = \frac{2}{\pi d}\frac{1}{1+(y/d)^2}, \quad 0 < y < \infty.$$

This is the Cauchy distribution restricted to $(0, \infty)$, and the mean is infinite.

2.23 $f_X(x) = \frac{1}{2}(1+x)$, $-1 < x < 1$



a) $Y = X^2$ $\quad X = g^{-1}(y) = \begin{cases} \sqrt{y} \\ -\sqrt{y} \end{cases}$, $\left|\frac{d}{dy} g^{-1}(y)\right| = \frac{1}{2\sqrt{y}}$

Thm 2.1.8

$$f_Y(y) = \frac{1}{2\sqrt{y}}\left[\frac{1}{2}(1+\sqrt{y}) + \frac{1}{2}(1-\sqrt{y})\right]$$

$$= \frac{1}{2\sqrt{y}}[1] = \boxed{\frac{1}{2\sqrt{y}}, \quad y \in (0,1)}$$

b) $E(Y) = \int_0^1 \frac{1}{2}\sqrt{y}\, dy = \frac{1}{2}\cdot\frac{2}{3} y^{3/2}\Big|_0^1 = 1/3$

$E(Y^2) = \int_0^1 \frac{1}{2} y^{3/2} dy = \frac{1}{2}\cdot\frac{2}{5} y^{5/2}\Big|_0^1 = \frac{1}{5}$

$\mathrm{Var}(Y) = E(Y^2) - (E(Y))^2 = \frac{1}{5} - \frac{1}{9} = 4/45$

8

$$F_Y(y) = P(Y \le y) = P(X(2-X) \le y)$$
$$= P(2x - x^2 \le y)$$
$$= P(x^2 - 2x \ge -y)$$
$$= P(x^2 - 2x + 1 \ge 1 - y)$$
$$= P((x-1)^2 \ge 1-y)$$
$$= P((X-1) \le -\sqrt{1-y}) + P((x-1) \ge \sqrt{1-y})$$
$$= P(X \le 1 - \sqrt{1-y}) + P(X \ge 1 + \sqrt{1-y})$$
$$= \int_0^{1-\sqrt{1-y}} \frac{x}{2} dx + \int_{1+\sqrt{1-y}}^{2} \frac{x}{2} dx$$
$$= 1 - \sqrt{1-y}$$

$$\boxed{f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{1}{2\sqrt{1-y}}, \quad 0 < y < 1}$$

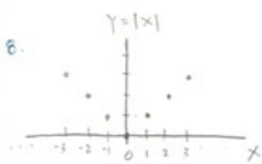
9

$g(x) = Y = 4 - x^3$

$\Rightarrow x = (4-y)^{1/3} \Rightarrow \frac{dx}{dy} = -\frac{1}{3}(4-y)^{-2/3}$

$$f_Y(y) = \frac{(4-y)^{1/3}}{2}\left|-\frac{1}{3}(4-y)^{-2/3}\right| = \boxed{\left(\frac{1}{6}\right)\frac{1}{(4-y)^{1/3}}, \quad -4 < y < 4 = f_Y(y)}$$

10 8.



for $Y = 1, 2, 3, \ldots$

$P(Y=y) = P(|X| = y) = P(X=-y) + P(X=y)$

$= \frac{1}{2n+1} + \frac{1}{2n+1} = \frac{2}{2n+1}$

for $Y=0$ $\quad P(Y=y) = P(X=y) = P(X=0)$

$$f_Y(y) = \begin{cases} \frac{2}{2n+1}, & y = 1, 2, 3, \ldots \\ \frac{1}{2n+1}, & y = 0 \\ 0 & \text{otherwise} \end{cases}$$

11.

3.16 a. Using integration by parts with, $u = t^\alpha$ and $dv = e^{-t}dt$, we obtain

$$\Gamma(\alpha+1) = \int_0^\infty t^{(\alpha+1)-1}e^{-t}dt = t^\alpha(-e^{-t})\Big|_0^\infty - \int_0^\infty \alpha t^{\alpha-1}(-e^{-t})dt = 0 + \alpha\Gamma(\alpha) = \alpha\Gamma(\alpha).$$

b. Making the change of variable $z = \sqrt{2t}$, i.e., $t = z^2/2$, we obtain

$$\Gamma(1/2) = \int_0^\infty t^{-1/2}e^{-t}dt = \int_0^\infty \frac{\sqrt{2}}{z}e^{-z^2/2}zdz = \sqrt{2}\int_0^\infty e^{-z^2/2}dz = \sqrt{2}\frac{\sqrt{\pi}}{\sqrt{2}} = \sqrt{\pi}.$$

where the penultimate equality uses (3.3.14).

12.

a. $\lim_{n\to\infty}\left(\frac{\theta n}{a+\theta n}\right) = \lim_{n\to\infty}\left(\frac{a+\theta n}{\theta n}\right)^{-\theta n} = \lim_{n\to\infty}\left(1+\frac{a}{\theta n}\right)^{-\theta n}$

$= \lim_{n\to\infty}\left[\left(1+\frac{a/\theta}{n}\right)^{n}\right]^{-\theta} = \left(e^{a/\theta}\right)^{-\theta} = \boxed{e^{-a}}$

b. $\lim_{N\to\infty}\left(\frac{N-x}{N}\right)^{N} = \left(1-\frac{x}{N}\right)^{N} = \boxed{e^{-x}}$

c. $\lim_{(n-m)\to\infty}\left(\frac{n-m}{n-m-K+x}\right)^{n-m} = \lim_{N\to\infty}\left[\frac{N}{N+(x-K)}\right]^{N}$

$= \lim_{N\to\infty}\left(\frac{N+(x-K)}{N}\right)^{-N} = \lim_{N\to\infty}\left[\left(1+\frac{x-K}{N}\right)^{N}\right]^{-1} = e^{(x-K)-1} = \boxed{e^{-(x+K)}}$